Tech-takers Team
Inbound area process automation

Ahmed EL-Badrawy (Biochemical Engineering)
Youssif Attia (Mechatronics Engineering)
Shahd Negm (Mechatronics Engineering)

UNIVERSITY of DEBRECEN



Table of Contents

Introduction
Challenge description
01

Solution
Proposed technical solution
02

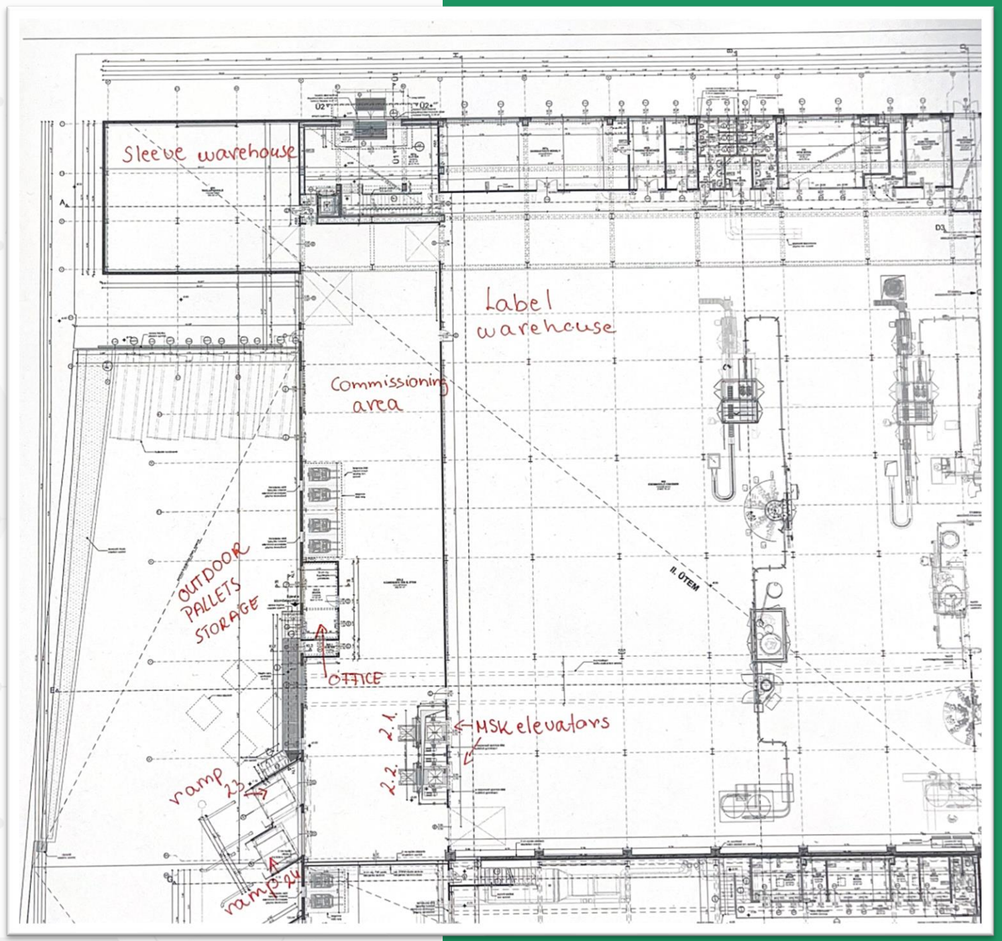
On-site implementation
Cost and Dimensions
03

Solution analysis
Results and Expected advantages
04

1. Introduction

1.1 Problem overview:

- Current inbound process is manually handled across multiple stages.
- Workers manage shipment document reception and manual SAP data entry.
- Pallets are unloaded from trucks using forklifts.
- SAP labels are printed and applied manually.
- Pallets are transferred to elevators and commissioning areas by workers.
- The workflow is fragmented and labour-intensive.
- Results in high labor costs and frequent process delays.
- Increases the risk of human error, particularly in data entry and labelling.



1.2 Problem in details:

The factory's manual processes create bottlenecks and inefficiencies. The system has round-the-clock staffing and data entry errors. Our proposed solution automates key steps.

Manual Operations

- Unloading pallets
- SAP document registration
- Label printing and application
- Pallet movement

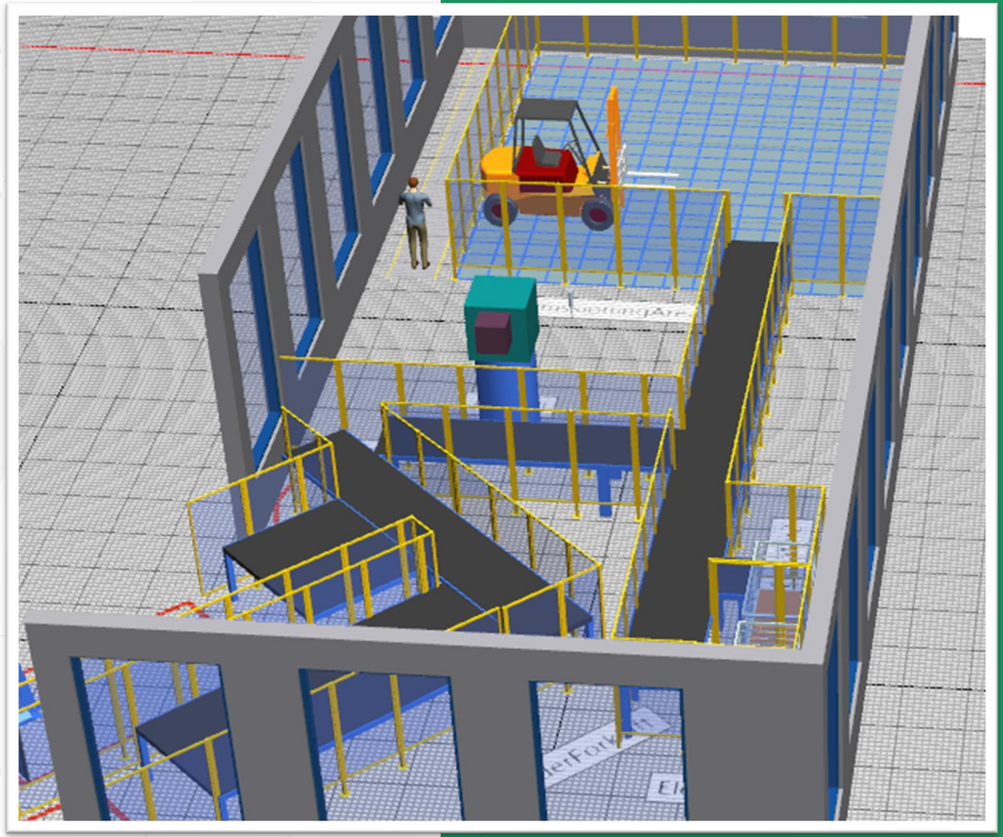
Consequences

- High labor costs
- Process delays
- Risk of human error

2. Solution

Automated slider:

The proposed system automates inbound logistics for maximum efficiency. The automated slider setup integrates directly with SAP. Auto-labeling and pallet orientation ensure precision.



Driver-Operated SAP integration

Streamlines data entry at the source.

Automated Unloading

Reduces manual labor and speeds up the process.

Smart Conveyor Transport

Provides consistent and safe pallet transfer.

2.1 SAP Integration via Driver Input:

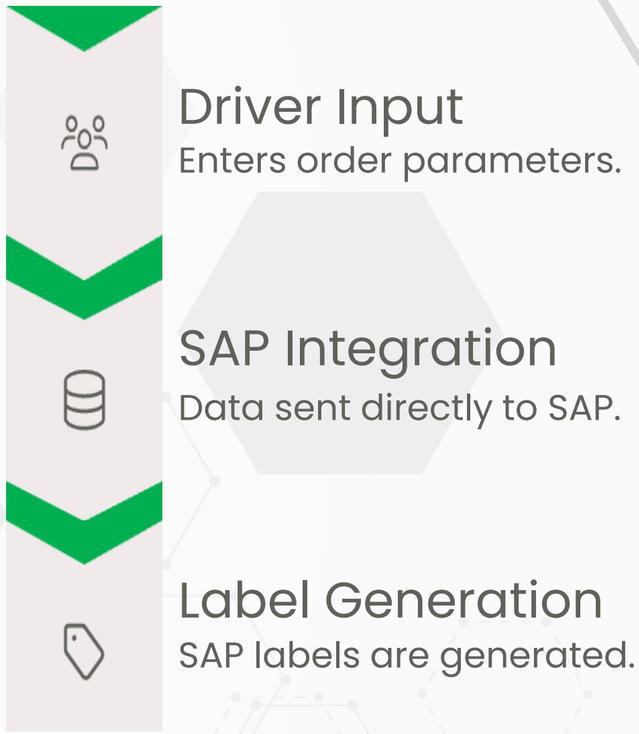
Our solution begins with digitizing the very first touchpoint in the inbound process. Upon truck arrival, the driver is directed to input four critical order parameters into a user-friendly digital interface:

1. **Number of order**
2. **Type of pallet**
3. **Weight of pallet**
4. **Type of material**

- Interface is directly connected to the factory's SAP system.
- Automatically processes entered data and generates SAP labels without administrative intervention.
- System identifies pallet type (EU or UK) and guides the driver to the appropriate unloading ramp.
- **One fixed ramp is designated for EU pallet trucks, another for UK pallet trucks.**
- Slider design varies by pallet configuration:

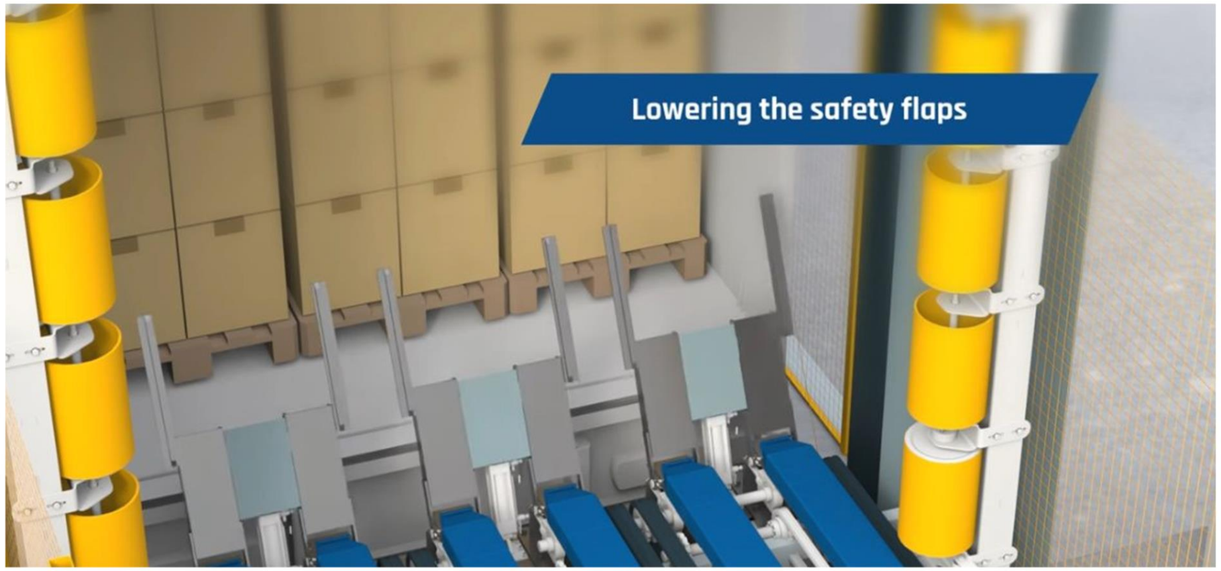
1. EU pallet trucks use a slider for three pallets per row.
2. UK pallet trucks use a slider for two pallets per row.

- Eliminates the need for manual handling and data entry by warehouse administrators.
- Reduces human error and improves overall efficiency.



2.2 Truck Dock Leveling System

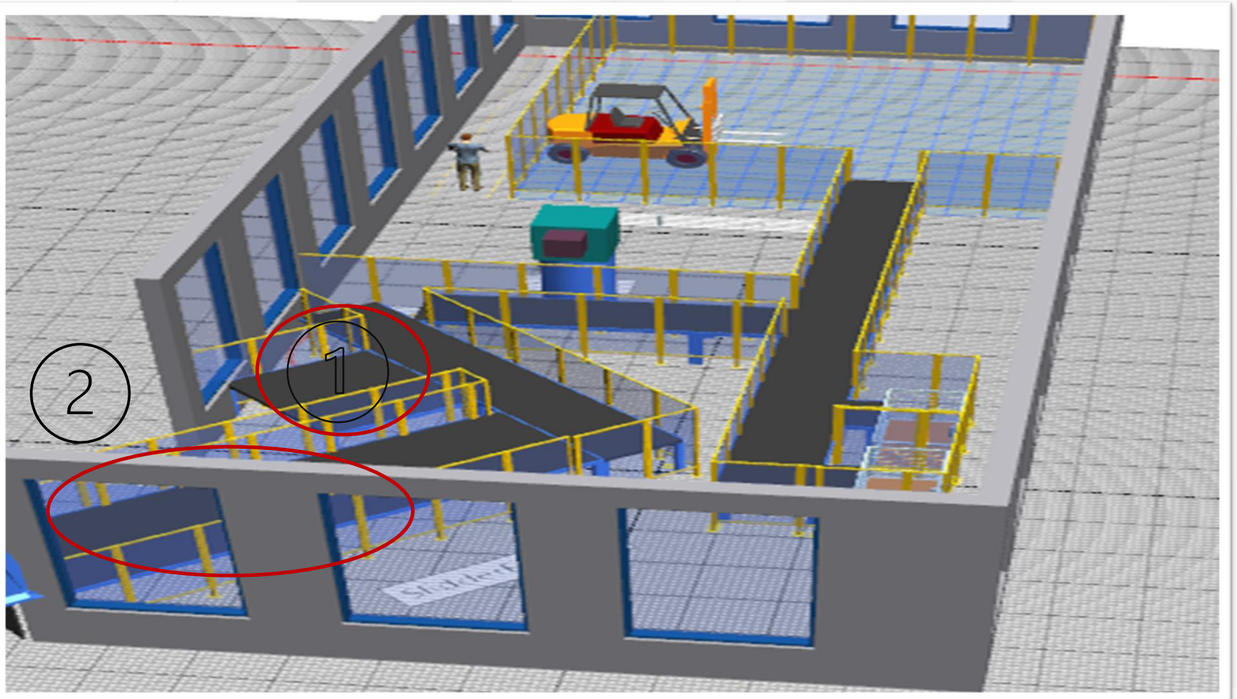
- Installed a levelling dock with safety flaps at the end of the slider.
- Ensures seamless physical interface between the truck and the slider system.
- Automatically adjusts to align with the height of the trailer floor.
- Provides a stable and consistent transition path for unloading.
- Slider and conveyor are installed at the same level as the elevator entrance.
- Guarantees accurate entry of pallets into the elevators.
- Eliminates height variances that could disrupt the unloading process.
- Ensures reliable and efficient operation of the slider system.



2.3 Slider Unloading

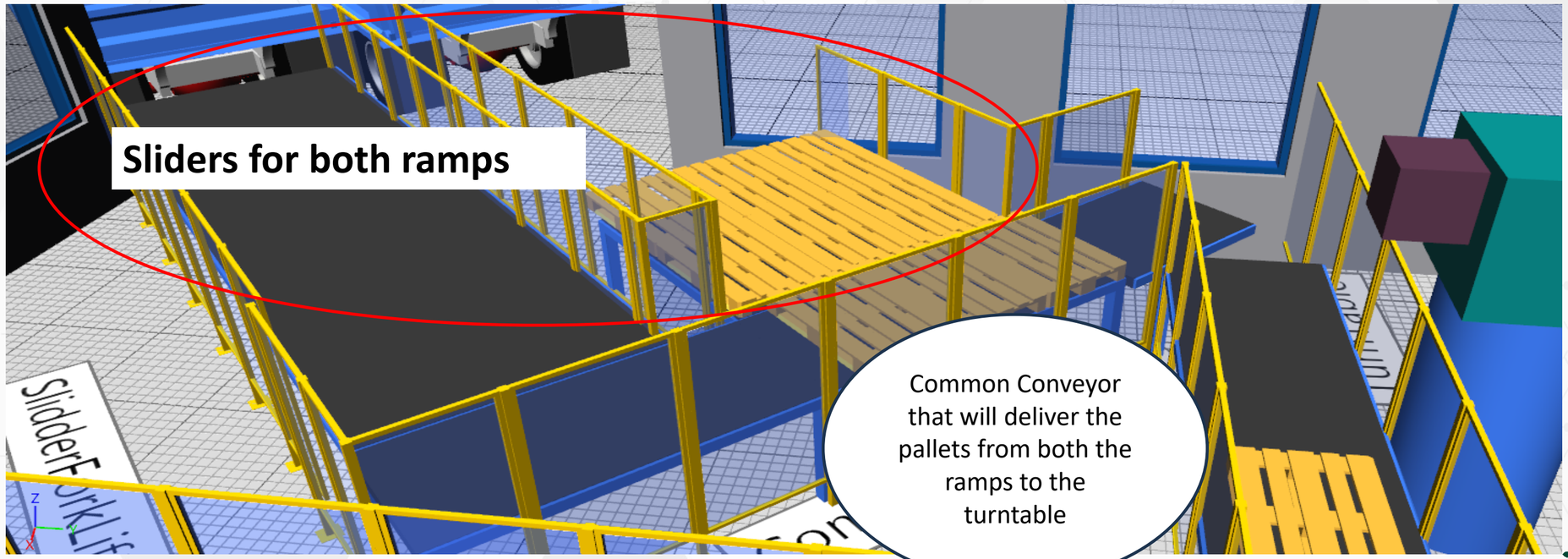
- Replaces forklifts with a stationary slider positioned in front of the ramp zone.
- Slider is foldable to fit within the available space.
- System autonomously unloads pallets from the truck in just three rounds.
- After each round, the conveyor begins delivering the pallets.
- Utilizes mechanical pushing and conveyor-based techniques for quick, safe extraction.
- Significantly reduces unloading time and manual labour.
- Enhances overall speed and efficiency of the unloading process.





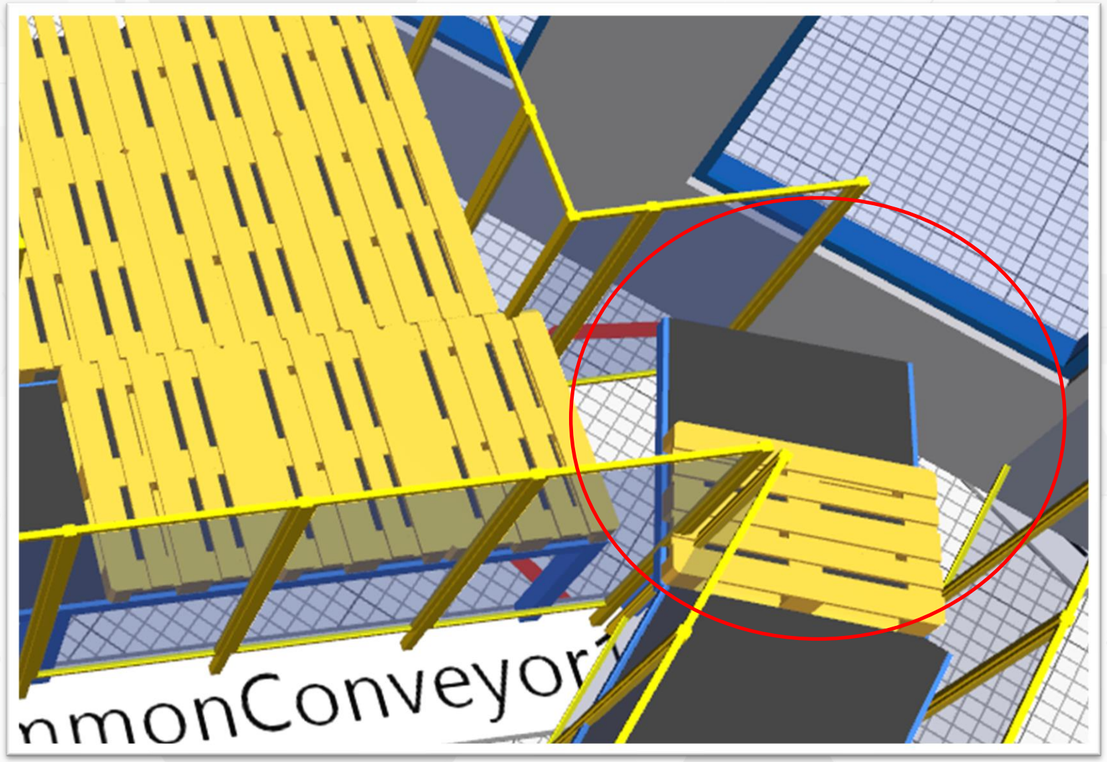
2.4 Conveyor Transport with Side Walls

- Unloaded pallets are placed directly onto a motorized conveyor system.
- Conveyor is equipped with protective side walls to maintain pallet stability.
- Side walls are crucial for handling varying pallet heights and weights.
- Prevents pallets from tipping over during movement through the warehouse.
- Enables continuous, hands-free transport from the unloading point.
- Directs pallets toward labeling and elevator areas efficiently.



2.5 Turntable for Adjusting the angled ramp

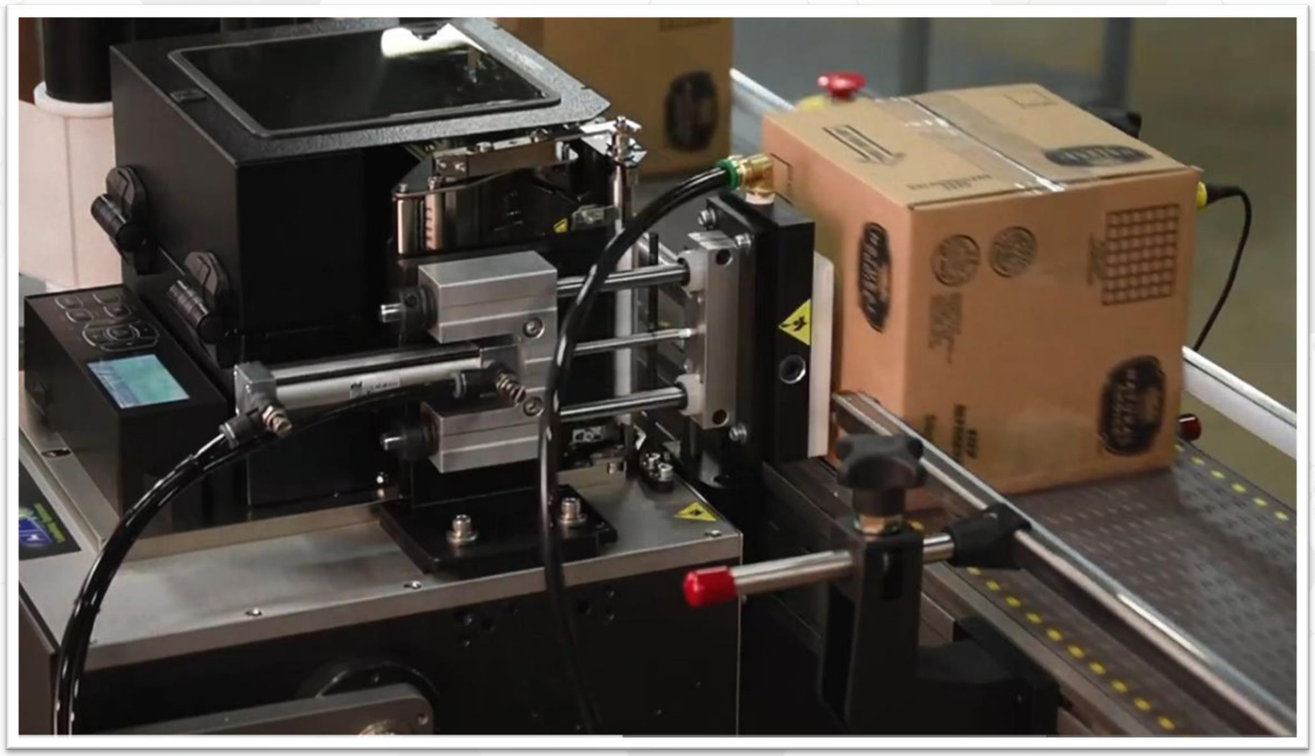
Upon reaching the turntable, the pallet is rotated 45° to align with the next conveyor. Once aligned, the rollers of both conveyors synchronize to ensure smooth and continuous pallet transfer.

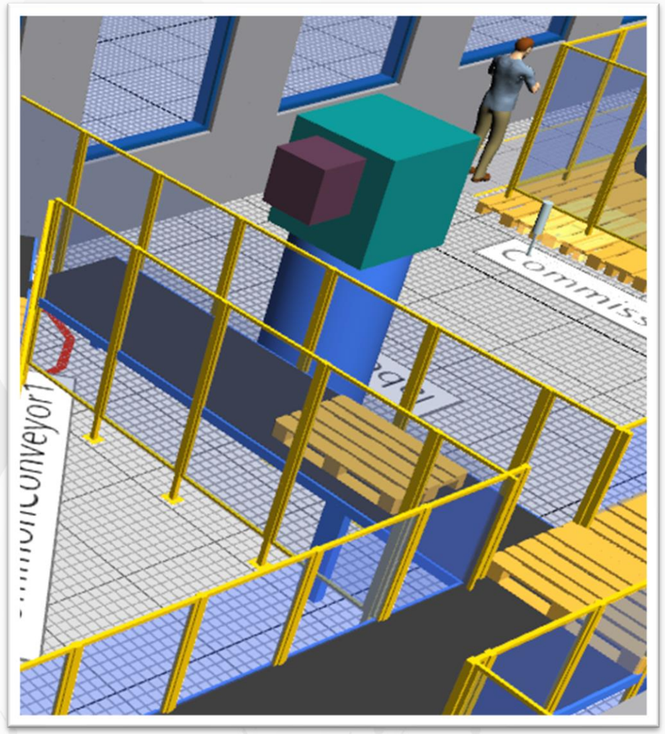




2.6 Automatic SAP Labeling Station

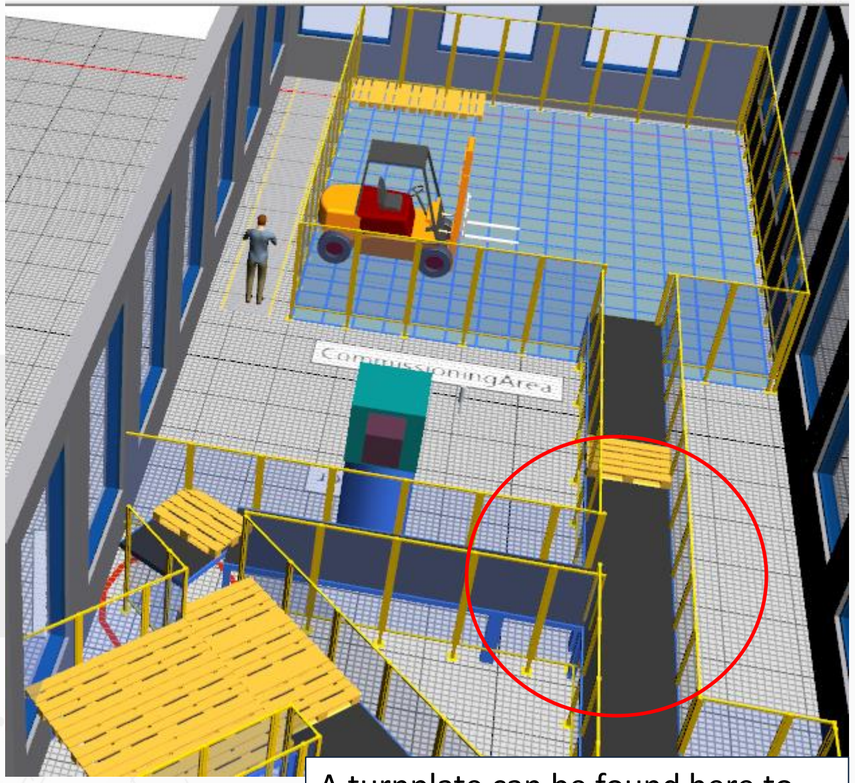
- A fixed labelling station is installed along the conveyor.
- Reads data from the initial truck check-in to generate the correct SAP label.
- Each pallet stops briefly at the station for automatic label application.
- Ensures consistent and precise label placement.
- Minimizes the risk of human error during labelling.
- Fully synchronized with the SAP system for accurate pallet identification.





2.7 Conveyor Extended to Commissioning Area

- After labelling and alignment, pallets are routed either to the elevator or a conveyor extension leading to the commissioning area.
- The extended conveyor section delivers both labels and glues to the designated Commissioning Area.
- Functions as a buffer zone for other material types when needed.
- Upon arrival at the commissioning area, Autonomous Mobile Robots (AMRs) handle the final internal logistics.
- The system uses driver-input material type to determine the pallet's route early in the process.
- Ensures synchronized routing and maximizes efficiency of the conveyor network.

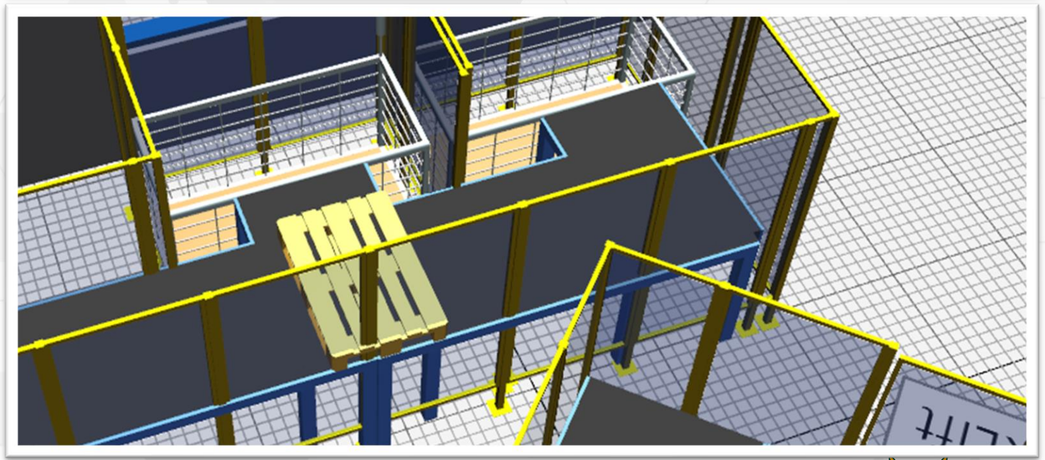


A turnplate can be found here to synchronous the conveyor rollers accordingly

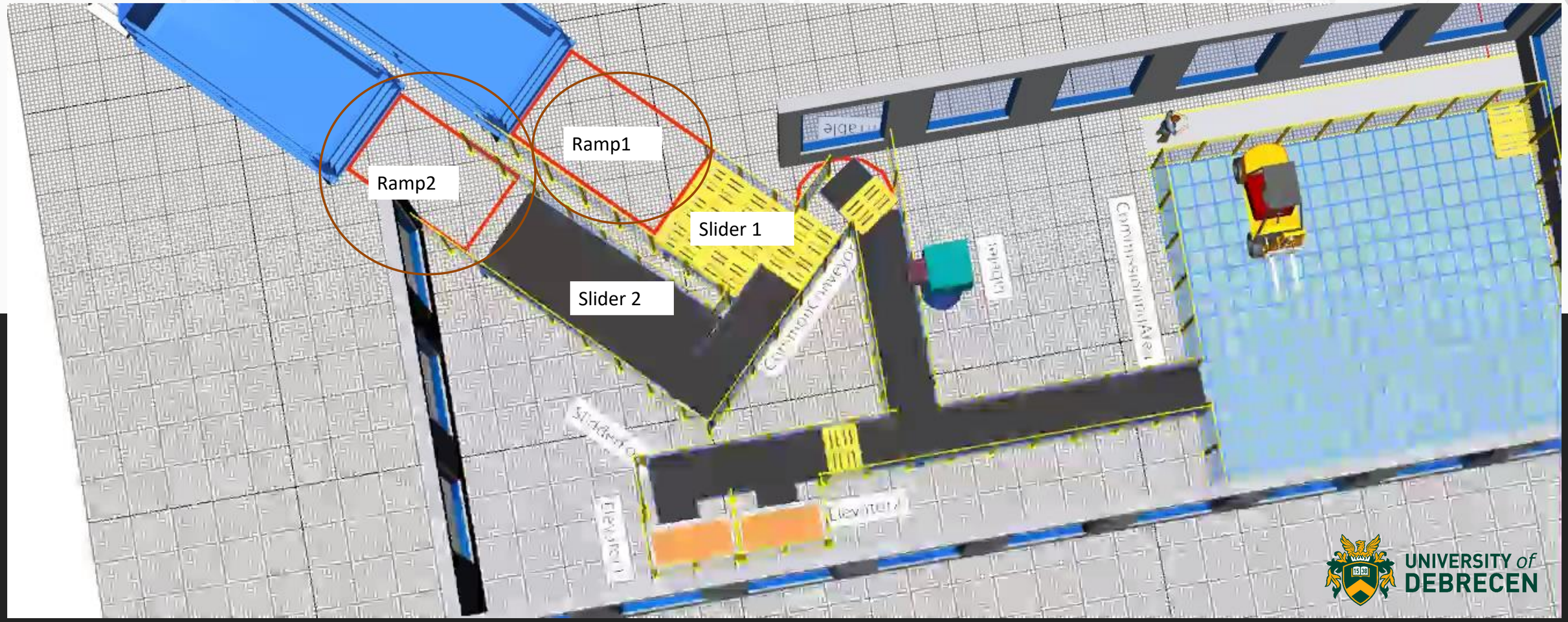
2.8 Direction-adjusting mechanism

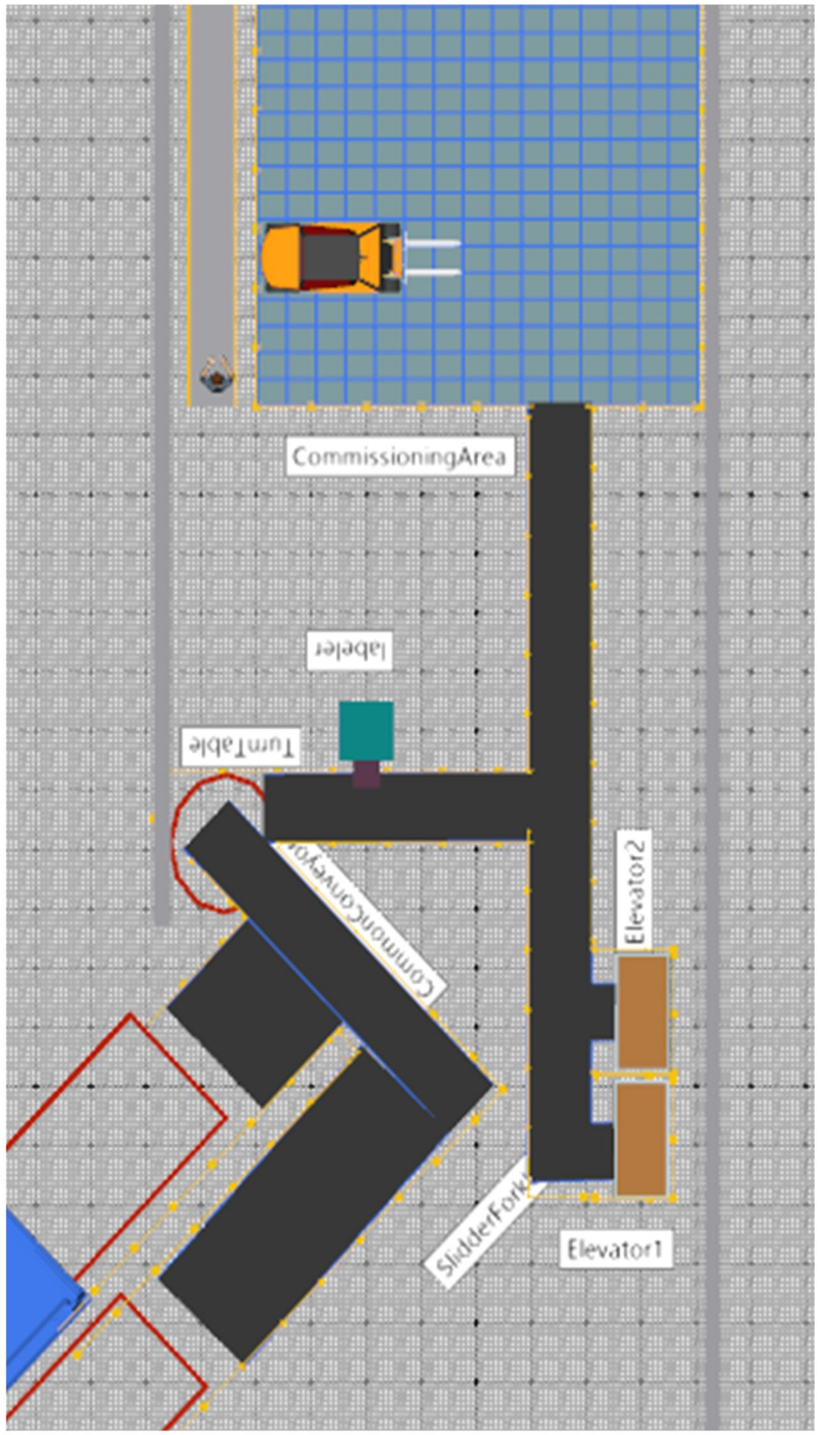
In front of each elevator, a direction-adjusting mechanism consisting of two perpendicular conveyor lines is integrated into the system. When a pallet arrives at the junction, sensors detect its presence, and the rollers of the primary conveyor are temporarily deactivated or lowered. Simultaneously, the perpendicular conveyor is activated, enabling the pallet to transfer laterally without rotation. This controlled transfer ensures precise alignment with the elevator entry and eliminates the need for manual repositioning.





2.9 Simulation of the whole system





3. On site implementation:

- Designed to operate within the constraints of the existing factory layout.
- No structural modifications required for truck ramps or elevators.
- Modular conveyor and slider systems adapt to various dock configurations.
- All components—SAP integration, pallet rotation, etc.—communicate seamlessly.
- Easily integrates with existing MES and AMR systems.
- Scalable and replicable for implementation in other warehouse areas.

3.1 Physical compatibility

- Conveyor system supports standard EU and UK pallet sizes.
- Designed to handle pallets up to 2100 mm in height and 750 kg in weight.
- All dimensions and load capacities are fully considered.
- Turning mechanism is spaced to allow complete pallet rotation.
- Ensures smooth and reliable operation for all incoming goods.

3.2 Estimated cost breakdown

This table breaks down the estimated costs. These costs include equipment, installation, and integration. It provides a transparent view of the investment needed for full automation.

Component	Estimated Cost (EUR)	Source
Driver SAP Terminal (App)	5000 setup fee +100/month	SAP
Automated slider System	15,000	WDX
Conveyor System (with side walls)	12,000	Bastian Solutions
Labeling Station (Print & Apply)	6,500	SATO EU distributors
Turntables	4,000	Rotary solutions
Sensors & Control Logic & PLC	4,000	HEITEC Hungary
Truck Dock Leveling System & Safety Flaps	4,500	KOPRON
Installation fees & Annual Maintenance	6,000	
Savings annually from workers only	>150,000	
Total Estimated Cost	59,000	

3.3 Cost comparison

Feature	Manual Process	Full AMR-Based Unloading	Our Hybrid Automated System
Labor Cost	High	Low	Very Low
Infrastructure Change	None	High	Medium
SAP Integration	Manual	Complex OCR	Simple Driver Input
Scalability	Limited	High	High
Accuracy	Low	Medium-High	High
Investment Cost	Low	Very High	Moderate

Our hybrid system offers superior cost-effectiveness compared to full automation. It balances infrastructure changes
and scalability. The hybrid automated system provides increased accuracy, reducing the chances of errors.

3.4 Cost efficiency summery

1

Reduced Staffing

Reduces staffing by 2–3 FTEs per shift. This leads to annual savings of €150,000 or more.

2

Modular Design

Modular design minimizes upfront costs compared to full automation solutions.

3

Lower Error Rates

The automated system significantly lowers error rates, reducing waste and returns.

4

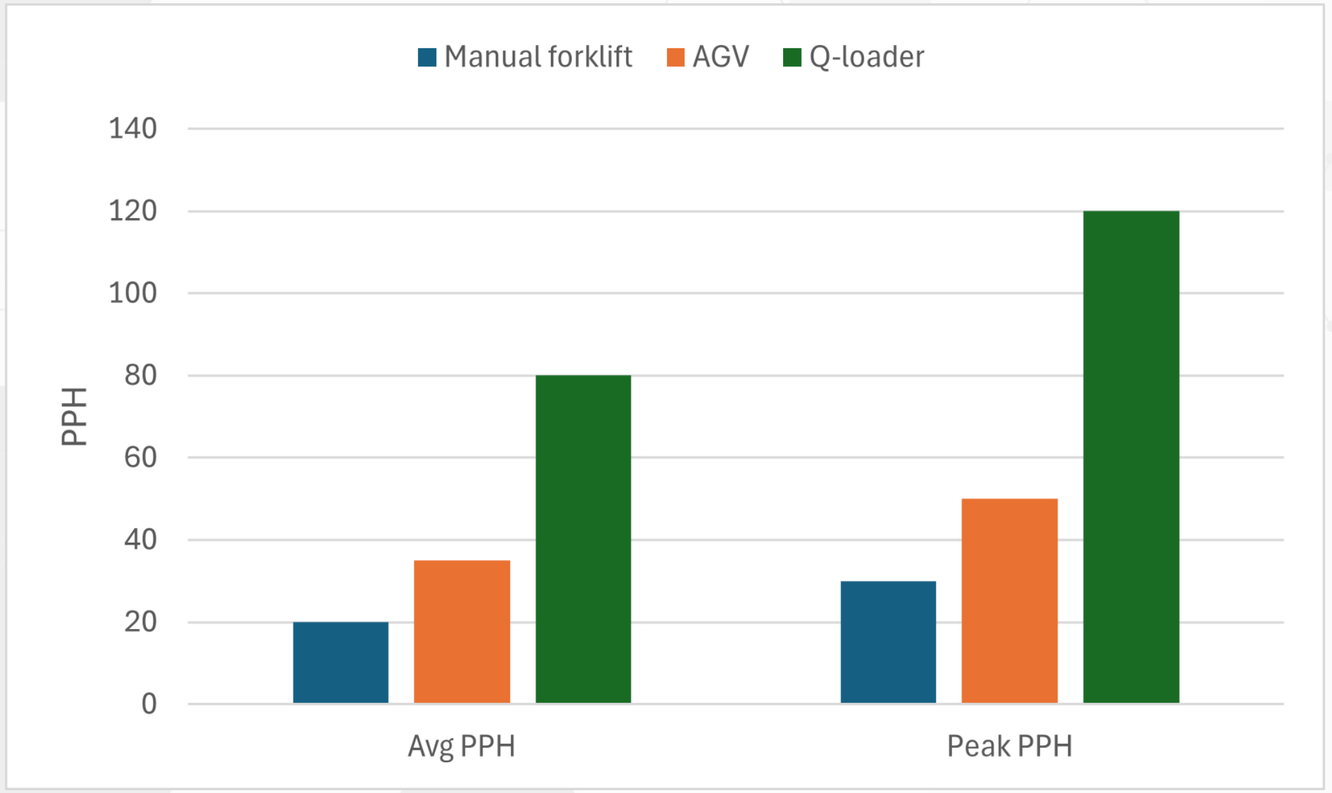
Seamless Integration

Easily integrates with existing SAP and AMR systems for maximum compatibility.

4. Solution Analysis

4.1 Time & Efficiency

By automating the unloading, labeling, and pallet handling process, the system drastically cuts down cycle times. Unloading that previously took up to 30–45 minutes can now be completed in under 10 minutes. The conveyor ensures smooth transitions between steps without manual intervention.



4.1 Time & Efficiency

Replacing forklift operations and manual SAP input reduces the number of full-time equivalents (FTEs) needed per shift by 2–3 workers. This leads to substantial labor cost savings over time. Additionally, the. automation of labeling and SAP registration reduces administrative mistakes that can disrupt inventory and production.

4.2 Compatibility & Flexibility

The modularity of the system means it can be easily adjusted or expanded to fit new layouts. Its integration with the existing SAP and AMR systems ensures that the factory benefits from end-to-end digital visibility and control.

System Flow Diagram

Truck Arrives

Driver enters order details.

Trailer Unloading

Pallets unloaded automatically.

Labeling Station

Labels are applied to pallets.

Elevator Handoff

Pallets are transferred for delivery.

SAP Label Generation

Labels are created automatically.

Conveyor Transport

Pallets are moved via conveyor.

Directional Adjustment

Pallets are aligned correctly.

SWOT Analysis for Automated Guided Vehicles

Strengths

- Works in existing layout.
- Easier integration.

Weaknesses

- Slower unloading.
- High initial cost.

Opportunities

- Versatile usage.
- Real-time traffic optimization.

Threats

- Software downtime.
- Interference with workers.

SWOT Analysis for Automated Trailer Unloading

Strengths

- Rapid unloading (2/3 rounds).
- Straight-line efficiency to elevator.
- Flexible transport to commissioning.

Weaknesses

- Significant mechanical setup needed.
- Less adaptable to layout changes.

Opportunities

- Standardize truck unloading.
- Reduce truck turnaround time.
- Enable full end-to-end automation.

Threats

- Driver cooperation for alignment.
- Ramp area constraints.

Conclusion

Our inbound automation concept introduces a comprehensive solution that addresses each pain point in the current manual process. From the moment a truck arrives, the process becomes digital, structured, and autonomous. By eliminating manual unloading and SAP tasks, aligning pallets accurately, and handing off to AMRs, this solution delivers a fully integrated, scalable, and future-ready system.

Balanced Automation

Combines smart mechanical design with digital integration.

Reduced Labor

Eliminates key manual tasks and lowers staffing requirements.

Scalable System

Supports future scalability to handle increased volumes.

Seamless Flow

Efficient, accurate, and ready to transform the inbound area.

UNIVERSITY of DEBRECEN

Thank You